

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (CL) Examination

March-April 2018

CL 762 Prestressed Concrete Structures

Date: 03.04.2018, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. Use of IS: 1343-2012 is permitted.
6. Assume density of concrete as 24 kN/m^3 unless otherwise stated.

SECTION – I

Q – 1 Choose the best alternative in the following

[07]

- (i) Select the incorrect statement.
 - (a) The loss of prestress is more in pre-tensioning system than in post-tensioning system
 - (b) Pre-tensioning system has greater certainty about its durability
 - (c) For heavy loads and large spans in buildings or bridges, post-tensioning system is cheaper than pre-tensioning system
 - (d) None of these
- (ii) The final deflection, due to all loads including the effects of temperature, creep and shrinkage and measured from the as-cast level of the supports of floors, roofs and all other horizontal members, should not normally exceed
 - (a) span/350
 - (b) span/300
 - (c) span/250
 - (d) span/200
- (iii) Loss of stress with time at constant strain in steel is called
 - (a) relaxation
 - (b) creep
 - (c) shrinkage
 - (d) ductility

- (iv) In a prestressed concrete beam subjected to prestress only
- (a) pressure line shifts from the cable line towards the top of beam
 - (b) pressure line coincides with cable line
 - (c) pressure line shifts from the cable line towards the soffit of beam
 - (d) none of the above
- (v) The prestress in the concrete at transfer is called
- (a) initial prestress
 - (b) partial prestress
 - (c) final prestress
 - (d) none of the above
- (vi) When the level of prestressing is such that no tensile stress is allowed in concrete under service loads, it is called
- (a) full prestressing
 - (b) limited prestressing
 - (c) partial prestressing
 - (d) none of the above
- (vii) A simply supported post-tensioned prestressed concrete beam of span L is prestressed by a straight tendon at a uniform eccentricity ' e ' below the centroidal axis. If magnitude of prestressing force is P and flexural rigidity of beam is EI , then maximum central deflection of the beam is
- (a) PeL^2/EI (downwards)
 - (b) $PeL^2/48EI$ (upwards)
 - (c) $PeL^2/8EI$ (upwards)
 - (d) $PeL^2/8EI$ (downwards)

Q – 2(a) Write the advantages of prestressed concrete. **[07]**

Q – 2(b) A rectangular concrete beam, 100 mm wide by 250 mm deep, spanning over 8m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm below the centroidal axis. The beam supports a live load of 1.5 kN/m. Calculate the resultant stress distribution for the central cross section of the beam. **[07]**

(OR)

Q – 2(a) Why high grade concrete and high tensile strength steel wires are used in prestressed concrete? **[07]**

Q – 2(b) A post-tensioned T girder has flanges 300×200 mm and web 150×500 mm. The effective depth of cross section is 500 mm. If area of prestressing steel is 200 mm^2 , characteristic strength of concrete is 40 MPa and of steel is 1500MPa, calculate moment of resistance of the section. [07]

Q – 3(a) A post-tensioned prestressed concrete beam of rectangular section, 200 mm wide and 400 mm deep, is simply supported over a span of 5m. The effective prestressing force in the cable is 500kN. The cable is parabolic with an eccentricity of 140 mm below the centroidal axis at the centre of span and concentric at supports. Find the cracking moment. Grade of concrete is 40 MPa. [07]

Q – 3(b) Enlist the losses of prestress in pre-tensioning and post-tensioning. [07]

(OR)

Q – 3 The mid span section of a composite T beam comprises a pretensioned beam, 300 mm wide and 900 mm deep and an in situ cast slab 900 mm wide and 150 mm deep. The effective prestressing force located 200 mm from the soffit of the beam is 2500kN. The moment due to the weight of the precast section is 300 kNm at mid span. After this is erected in place, the top slab is cast producing a moment of 150 kNm at mid span. After the slab concrete is hardened, the composite section is to carry a maximum live load moment of 750kNm. Compute the resultant final stresses at [14]

- (a) the top of slab, and
- (b) the top and bottom of the precast section

SECTION – II

Q – 4 Choose the best alternative in the following [07]

(i) The profile of the centroid of the tendon is parabolic with central dip h . Effective prestressing force is P and span L . What is the equivalent upward acting uniform load?

- | | |
|---------------|---------------|
| (a) $8hL/P$ | (b) $8hP/L^2$ |
| (c) $8h^2L/P$ | (d) $8h^2P/L$ |

(ii) Stress at the bottom edge of a simply supported beam corresponding to the cracking moment is called

- (a) modulus of rupture
- (b) modulus of elasticity
- (c) characteristic compressive strength
- (d) ultimate tensile strength

- (iii) As per IS:1343-2012, at the time of initial tensioning the maximum tensile stress in tendon immediately behind the anchorage shall not exceed
- (a) 50% of the ultimate tensile strength of the wire or bar or strand
 - (b) 76% of the ultimate tensile strength of the wire or bar or strand
 - (c) 40% of the ultimate tensile strength of the wire or bar or strand
 - (d) 60% of the ultimate tensile strength of the wire or bar or strand
- (iv) The loss of prestress due to shrinkage of concrete is the product of
- (a) modular ratio and percentage of steel
 - (b) modulus of elasticity of concrete and shrinkage of concrete
 - (c) modulus of elasticity of steel and shrinkage of concrete
 - (d) modular ratio and modulus of elasticity of steel
- (v) Which one of the following method is employed to manufacture pre-stressed concrete sleepers for the railways?
- (a) Post-tensioning
 - (b) Pre-tensioning
 - (c) Pre-tensioning followed by post-tensioning
 - (d) Partial prestressing
- (vi) In pretensioning system
- (a) the member is cast first followed by tensioning of high tensile wires
 - (b) the member is prestressed by external anchors
 - (c) the high tensile wires are tensioned before placing concrete in forms
 - (d) None of the above
- (vii) In composite construction, prestressed elements are used advantageously in the
- (a) compression zone
 - (b) shear zone
 - (c) tension zone
 - (d) none of the above

Q – 5(a) A prestressed concrete beam of rectangular section, 150 mm wide and 300 mm deep, is simply supported over a span of 8m. The effective prestressing force in the cable is 200kN. The cable is parabolic with an eccentricity of 100 mm below the centroidal axis at the centre of span and concentric at supports. The beam supports a uniformly distributed load of 15kN/m including self-weight. Find the shear resistance of uncracked section at the support. Grade of concrete is 45MPa. **[07]**

- Q – 5(b)** A rectangular concrete beam of width 120 mm and depth 200 mm is prestressed by pre-tensioning to a force of 150kN at an eccentricity of 20mm below the centroidal axis. The cross sectional area of the pre-stressing steel is 187.5mm^2 . Take modulus of elasticity of steel and concrete as 2.1×10^5 MPa and 3×10^4 MPa respectively. Determine the percentage loss of stress in the pre-stressing steel due to elastic deformation of concrete. **[07]**

(OR)

- Q – 5** A prestressed concrete beam of rectangular section, 100 mm wide and 200 mm deep, is to be designed to support two imposed loads of 5 kN each located at one-third points over a span of 3m. If there is to be no tensile stress in the concrete at transfer and service loads, calculate the minimum prestressing force and the corresponding eccentricity. Loss ratio = 0.8 **[14]**
- Q – 6(a)** Using moment area method, derive the equation of deflection for simply supported prestressed concrete beam having parabolic cable (with eccentric anchors). **[07]**
- Q – 6(b)** Why it is necessary to control the deflection. Enlist the factors influencing the deflection of prestressed concrete members. **[07]**

(OR)

- Q – 6** A simply supported prestressed concrete beam spanning over 6 m is of rectangular section, 150 mm wide and 300 mm deep. The beam is prestressed by a parabolic cable having an eccentricity of 75 mm below the centroidal axis at the center of span and an eccentricity of 25 mm above the centroidal axis at the support sections. The initial force in the cable is 400 kN. The beam supports a concentrated load of 10 kN at mid-span. The modulus of elasticity of concrete is 38 kN/mm^2 . **[14]**
- (i) Neglecting losses of prestress, estimate the short-term deflection due to (Prestress + self-weight) and
- (ii) Allowing for 20% loss in prestress, estimate the long-term deflection due to (Prestress + self-weight + live load), assuming creep coefficient as 1.80.
